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Olist Marketplace Performance & Customer Experience (2016–2018)

1. Project Overview

This project analyzes the performance of the **Olist** Brazilian e-commerce marketplace using two years of real transaction data (September 2016 – October 2018). The goal is to understand:

- How the marketplace has performed over time (orders, revenue, review scores)
- How delivery performance impacts customer satisfaction
- How results vary by geography, product category and payment behavior
- What factors are most strongly associated with low review scores
- What actions Olist should prioritize to improve the customer experience

2. Dataset Summary

Source: Olist Brazilian E-Commerce Public Dataset (Kaggle)

Tables used:

- orders – 99,441 rows (one row per order)
- order_items – 112,650 rows (one row per item)
- order_payments – 103,886 rows (one row per payment record)
- order_reviews – 100,000 rows (one row per review)
- customers – 99,441 rows (order-level customer IDs)
- products – 32,951 rows (one row per product)
- sellers – 3,095 rows (one row per seller)
- geolocation and category_translation as supporting tables

Key fields used:

- **Orders & Delivery:** order status, purchase timestamp, delivery timestamps, estimated delivery date
- **Customers & Sellers:** city, state
- **Products:** product category
- **Payments:** payment type, payment value, installments
- **Reviews:** review_score, review creation and answer timestamps

Major data quality characteristics:

- Non-delivered / cancelled orders naturally lack delivery timestamps
- Many reviews have a numeric score but no text comment
- A small number of products are missing category or dimension fields

3. Data Preparation and EDA Using Python

I used Python (Pandas) in Google Colab for data preparation and initial exploration.

3.1 Data Loading and Type Handling

- Loaded all nine CSV files into Pandas DataFrames.
- Standardized data types:
 - Converted IDs (order_id, customer_id, product_id, seller_id) to string
 - Parsed all timestamp columns to datetime
 - Cast prices, freight, payment values and dimensions to numeric types

3.2 Feature Engineering on Orders

From the orders table, I engineered delivery-related features:

- delivery_time_days
= (order_delivered_customer_date - order_purchase_timestamp) in days
- delivery_delay_days
= (order_delivered_customer_date - order_estimated_delivery_date) in days
 - < 0 → delivered before estimate (early/on-time)
 - > 0 → delivered after estimate (late)
- is_late
= 1 if delivery_delay_days > 0, else 0

In addition, I created simple missing-value flags for delivery timestamps so that non-completed orders could be identified but not confused with data errors.

3.3 Loading to MySQL

The cleaned tables were loaded into a MySQL database (`olist_db`) using [SQLAlchemy's `to\_sql`](#).

MySQL was then used to write structured queries for each business question.

4. SQL-Based Analysis

I used SQL in MySQL Workbench to answer the core analytical questions.

4.1 Marketplace Performance Over Time

Question: How have order volume, revenue, and review scores trended over time?

	product_category_name	orders_count	total_revenue	avg_price_per_item	avg_review_score
▶	moveis_escritorio	1254	269418.0999999997	160.55905840285877	3.5083
	fashion_roupa_masculina	106	10452.329999999982	83.61863999999986	3.7360
	telefonos_fixa	212	55355.19999999995	216.23124999999982	3.7461
	audio	348	50620.500000000124	139.45041322314083	3.8127
	casa_conforto	392	58265.150000000256	134.873032407408	3.8472
	cama_mesa_banho	9272	1037326.58000000711	93.37713385543893	3.8978
	moveis_sala	414	67861.4000000001	135.99478957915852	3.9178
	unknown	1392	171438.58999999793	110.96348867313782	3.9217
	moveis_decoracao	6307	718744.7700000417	87.22630703884	3.9346
	casa_construcao	483	81527.41999999982	136.56184254606336	3.9464
	construcao_ferramentas...	159	38773.22000000004	211.87551912568327	3.9508
	artigos_de_natal	125	8737.839999999987	58.252266666666586	3.9533
	informatica_acessorios	6530	896324.1800000347	116.26983785186597	3.9765
	telefonos	4093	310004.1099999896	69.91522552999315	3.9871
	climatizacao	246	53323.560000000056	184.5105882352943	4.0000
	construcao_ferramentas...	242	40185.00000000004	133.0629139072849	4.0066
	moveis_cozinha_area_d...	241	45531.980000000076	166.17510948905138	4.0182
	fashion_underwear_e...	117	9305.949999999988	73.27519685039361	4.0236
	tablets_impressao_imagem	79	7528.409999999994	90.70373493975897	4.0482

Counting.....

4.2 Order Status Breakdown Over Time

Question: What share of orders are delivered vs. cancelled or still in progress each month?

	yearmonth	order_status	total_orders
▶	2016-09	canceled	2
	2016-09	delivered	1
	2016-09	shipped	1
	2016-10	canceled	24
	2016-10	delivered	265
	2016-10	invoiced	18
	2016-10	processing	2
	2016-10	shipped	8
	2016-10	unavailable	7
	2016-12	delivered	1
	2017-01	canceled	3
	2017-01	delivered	750
	2017-01	invoiced	12
	2017-01	processing	9
	2017-01	shipped	16
	2017-01	unavailable	10
	2017-02	approved	1
	2017-02	canceled	17
	2017-02	delivered	1653
	2017-02	invoiced	11

Running Row Counting....

4.3 Delivery Status vs. Customer Satisfaction

Question : Do late deliveries receive lower review scores than on-time deliveries?

	delivery_status	total_orders	avg_review_score
▶	On_time_or_early	89150	4.2831
	Late	7865	2.5465

4.4 State-Level Delivery Performance and Ratings

Question : Which customer states experience longer delivery times and lower review scores?

Note on delivery_delay_days (negative values)

The field `delivery_delay_days` is calculated as:

$$\text{delivery_delay_days} = \text{actual_delivery_date} - \text{estimated_delivery_date}$$

Because of this definition:

- **Negative values** mean the order was delivered **earlier than the estimated date** (early/on-time).

- **Zero** means delivery exactly on the estimated date.
- **Positive values** mean the order was delivered **later than the estimate** (late).

	customer_state	total_orders	avg_delivery_time_days	avg_delay_days	avg_review_score
▶	AL	401	24.53605263461716	-8.039265551399284	3.8204
	MA	721	21.521970112883324	-8.959156922612626	3.8197
	SE	335	21.51978828081814	-9.328641480099503	3.8985
	ES	2004	15.798111496913602	-9.800670932440646	4.0509
	BA	3274	19.354657415891836	-10.08971061279666	3.9096
	CE	1282	21.268040370225865	-10.123478948185694	3.9314
	MS	712	15.629537654754468	-10.347081447669568	4.1657
	SP	40715	8.765187065105078	-10.389201178492565	4.2345
	PI	477	19.45276130231385	-10.620075049499187	3.9811
	SC	3560	14.975162014279006	-10.784210345791697	4.1146
	RJ	12420	15.318877192737183	-11.046055835122534	3.9452
	DF	2099	12.996125309891799	-11.314706082481964	4.1253
	TO	274	17.65806260982698	-11.439737724722898	4.1569
	GO	1974	15.604079321691229	-11.49939978399754	4.0881
	MG	11423	12.011947980601201	-12.55461235717576	4.1799
	PR	4942	12.000591158362964	-12.608206557174336	4.2272
	PB	518	20.403992563992585	-12.615224599599593	4.0637
	PE	1606	18.429756685000676	-12.663140004497025	4.0691
	RN	476	19.263879235722065	-12.97316599556489	4.1303

Counting.....

4.5 Product Category Performance

Question : Which categories drive volume and revenue, and how do their review scores compare to the platform average?

	product_category_name	orders_count	total_revenue	avg_price_per_item	avg_review_score
►	moveis_escritorio	1254	269418.0999999997	160.55905840285877	3.5083
	fashion_roupa_masculina	106	10452.329999999982	83.61863999999986	3.7360
	telefonos_fixa	212	55355.19999999995	216.23124999999982	3.7461
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Counting....

4.6 Payment Behavior

Question : *How do different payment types relate to order volume and average order value?*

Note on `payment_type = 'not_defined'`

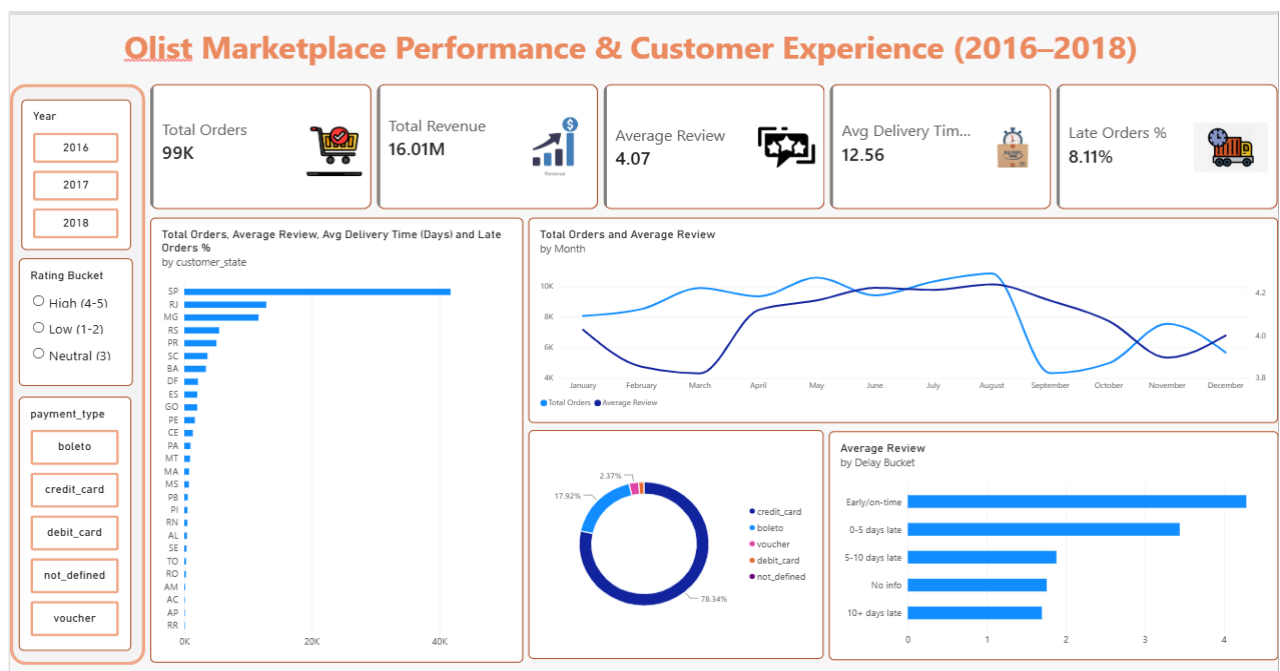
A small share of orders have `payment_type = 'not_defined'`. These are orders where the payment method was not properly recorded in the source system (for example, missing or inconsistent entries). In the analysis, they are included for completeness in overall order and revenue numbers, but they are **not interpreted as a real payment method**. Business decisions about payment behavior should focus on the defined types (credit card, boleto, debit card, voucher, etc.), not on this technical category.

	payment_type	orders_count	avg_order_value
▶	boleto	19784	145.0344354023444
	credit_card	76505	167.174986782986
	debit_card	1528	142.66763243950294
	not_defined	3	0
	voucher	3866	141.309733333333

5. Interactive Dashboard in Power BI

To bring the SQL and Python analysis together for business users, I designed a **single-page interactive Power BI dashboard** titled:

“Olist Marketplace Performance & Customer Experience (2016–2018)”



6. Business Recommendations

Based on the combined Python, SQL and Power BI analysis, my recommendations focus on the factors that most strongly affect customer experience and scalable growth.

- **Delivery Performance** – Reduce late orders, especially 10+ days late, with stricter SLAs and tracking.
- **ETA Accuracy** – Re-calibrate estimated delivery dates by region so promises match actual performance.
- **Seller Management** – Monitor seller-level delay and ratings; coach or penalize persistent low performers.
- **Category Quality** – Review low-rating categories for product/description issues; promote high-rating categories.
- **Logistics by Region** – Improve routes/carriers for states with long delivery times and weak reviews.
- **Payment Strategy** – Maintain focus on credit-card flow; use other payment types only for tactical promotions.